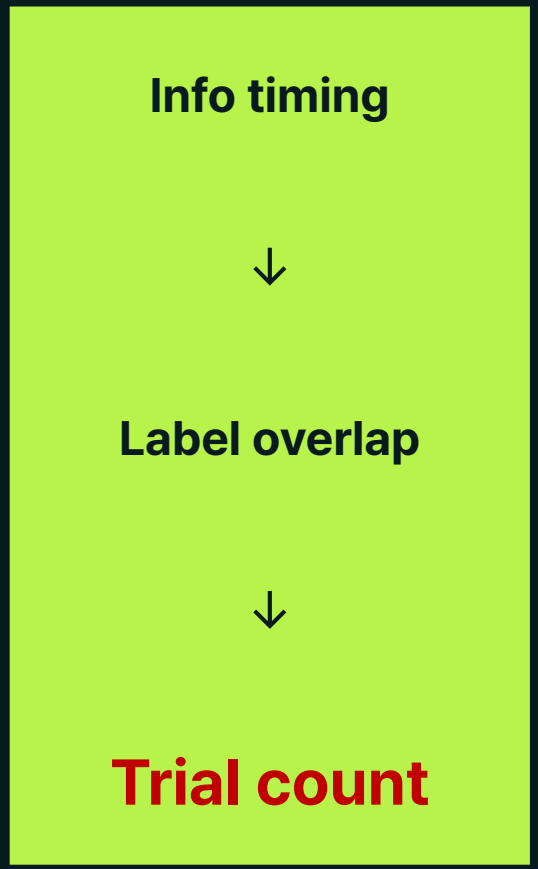
Honest Backtesting
Not Fooled by the Past

Does a strategy that earned well really work?



Financial ML · Week 10

This week runs in four steps

From basics to core concepts, cases, then summary

01	**Basics**	**Tools to read a backtest** Returns, train/test, overfitting
02	**Leakage, overlap**	**Use only what was known then** Common errors, purging, walk-forward
03	**Luck, multiple tests**	**Many trials yield lucky winners** Sharpe ratio, best Sharpe by luck, DSR
04	**Cases, summary**	**What actually happened** Decay, survivorship, factor zoo

Revising against the test raises scores, not skill

01

Basics

Concepts to know before reading backtest results

Backtest metrics use three ingredients

Return, excess return, volatility: all metrics here come from these

RETURN

Return

- The % price change over a period.

EXCESS RETURN

Excess return

- Return earned above the risk-free rate.

VOLATILITY

Volatility

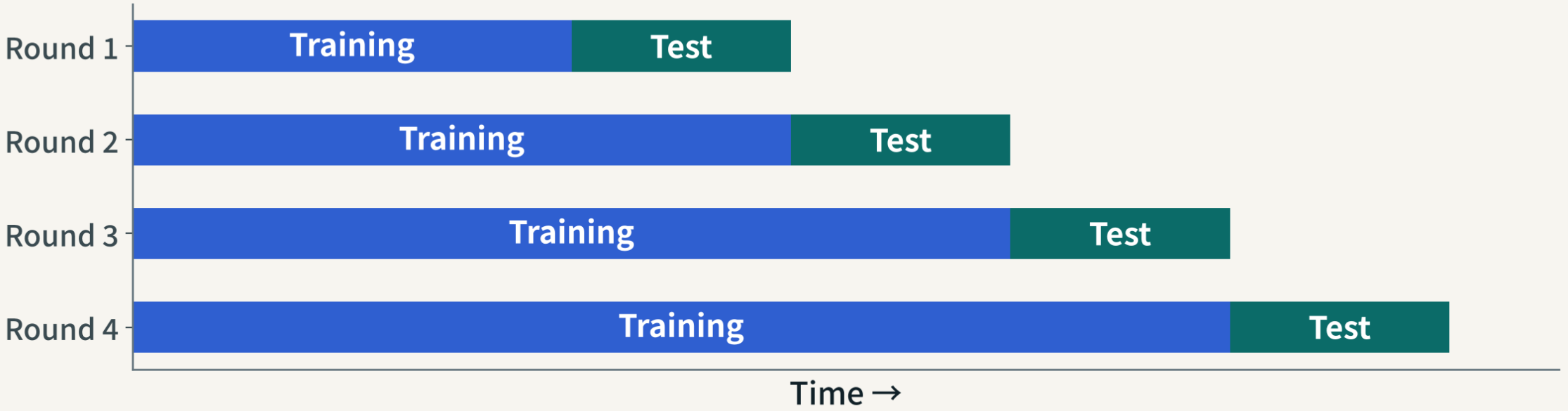
- How much returns swing around the mean (standard deviation).

Sharpe ratio is excess return divided by volatility

Mixing daily, monthly, annual units breaks numbers

Validation learns from the past, tests on the future

How training, test and out-of-sample periods relate



※ Each round trains on the past only and tests on the next period (expanding window)

The test period must be a future the model never saw

In-sample results cannot separate skill from luck

Overfitting is memorizing even past noise

Overfitting: fits training data, fails on new data

SIGNAL

Signal

- A pattern repeated across periods.
- It may persist in the future.

What we want to learn

NOISE

Noise

- A fluke of that period only.
- It vanishes in the future.

Not to be memorized

Complex models and many trials make noise look like signal

02

Leakage · Overlap · Luck

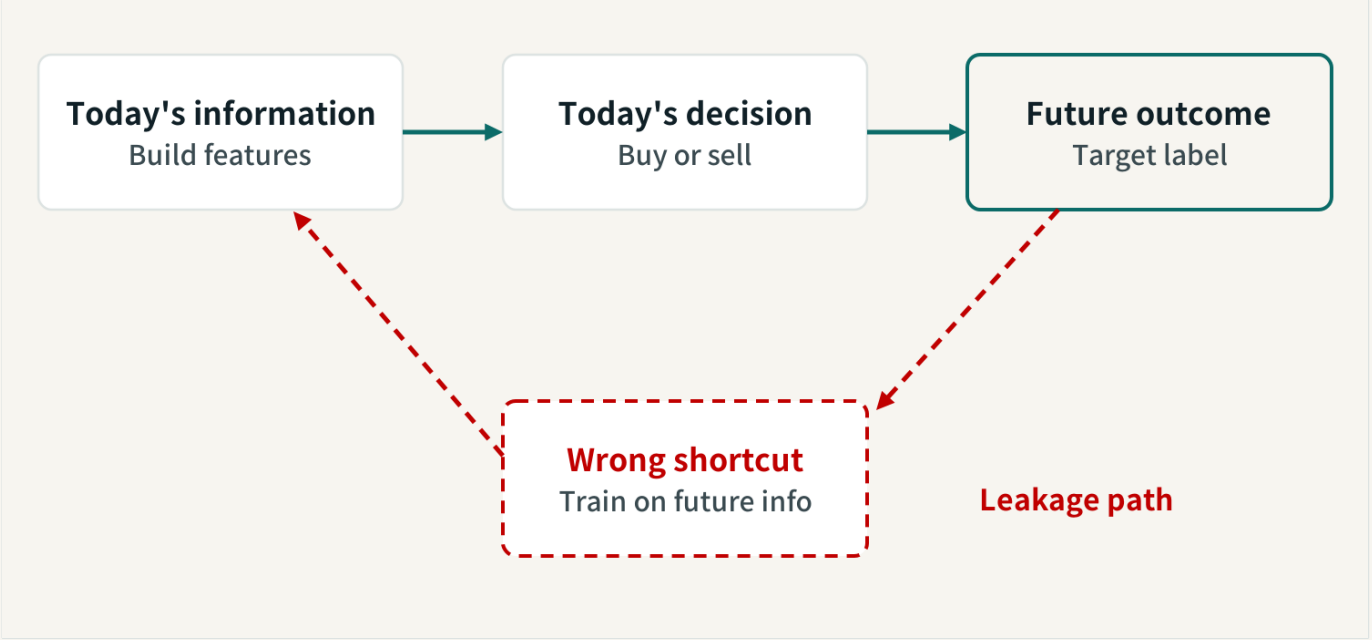
Source concepts: info timing, label spans, trials

A backtest may use only what was known at decision time

A backtest is a mock test replaying past data in time order

Look-ahead leakage

- Using earnings released after the close in that morning's trade peeks ahead.
- Standardizing with full-period mean and SD, then splitting, also leaks.



Fit scaling on training data only

Check information timing before adding model complexity

Four common mistakes all inflate performance

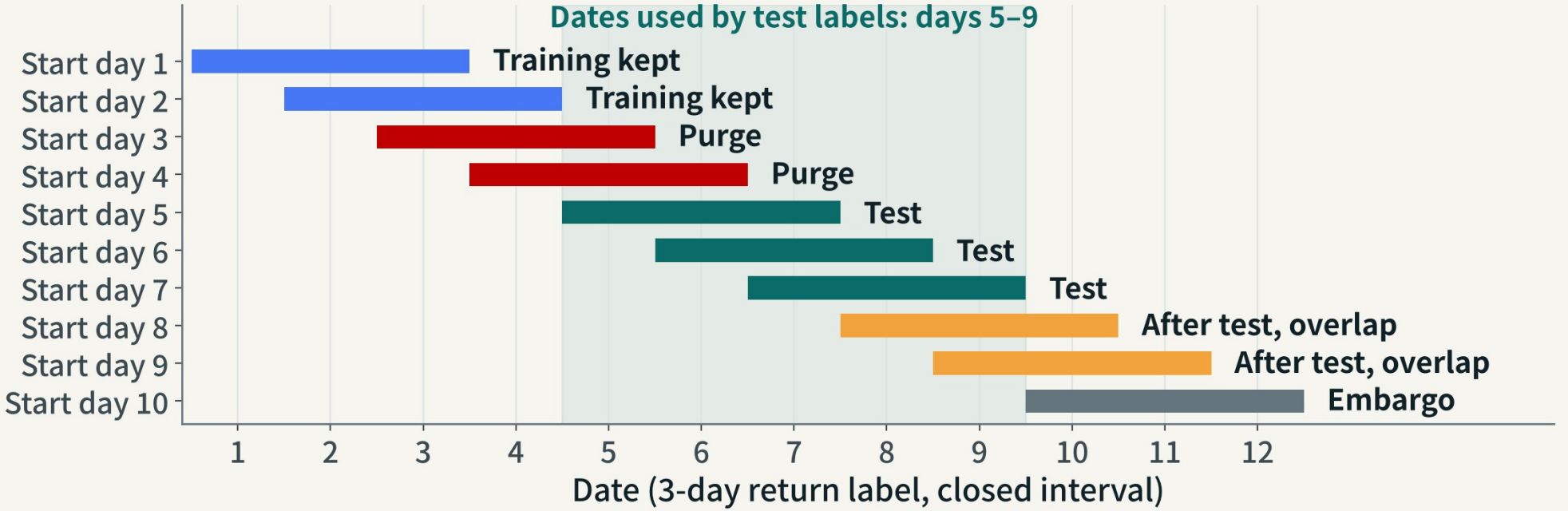
Each row: why the mistake inflates the score, and the fix

Common mistake	Why it inflates results	How to fix it
Shuffling time-series data	Future regimes help fit the past	Train on past, test on future
Surviving stocks only	Delisted failures drop out	Use stocks investable then
Preprocessing all data	Test-period traits leak into training	Fit within each training window
Revising after tests	The test set becomes training data	Split dev validation and final test

Revising against the test raises scores, not skill

Split by date, label windows can still overlap

3-day return labels, test starts days 5–7, closed intervals with endpoints



Test labels span days 5–9, overlapping training samples from days 3–4

Compare each sample's full window, not its start date

Purging and time order are separate constraints

'Purged CV' does not mean 'always train on the past only'

PURGING

Purging

- Drop training samples overlapping test info or label windows.

EMBARGO

Embargo

- Also drop some samples after the test window.
- Set its length to the data and info window.

WALK-FORWARD

Walk-forward

- Train on the past, test on later periods.
- Replicates the time order of live trading.

Some Purged CV also trains on data after the test window

※ Source note: the p.13 date example is simplified around start dates; do not generalize

Sharpe ratio is return divided by fluctuation

Compute the Sharpe ratio in one consistent period unit

Definition

$$SR = \frac{\text{mean excess return}}{\text{std. dev. of excess return}}$$

E.g. monthly mean excess return 1%, monthly vol. 2%

$$\text{Monthly Sharpe } \frac{1\%}{2\%} = 0.5 \implies \text{annualized } 0.5\sqrt{12} \approx 1.73$$

Annualizing assumes independent, stable variance. Do not use it as is under strong autocorrelation

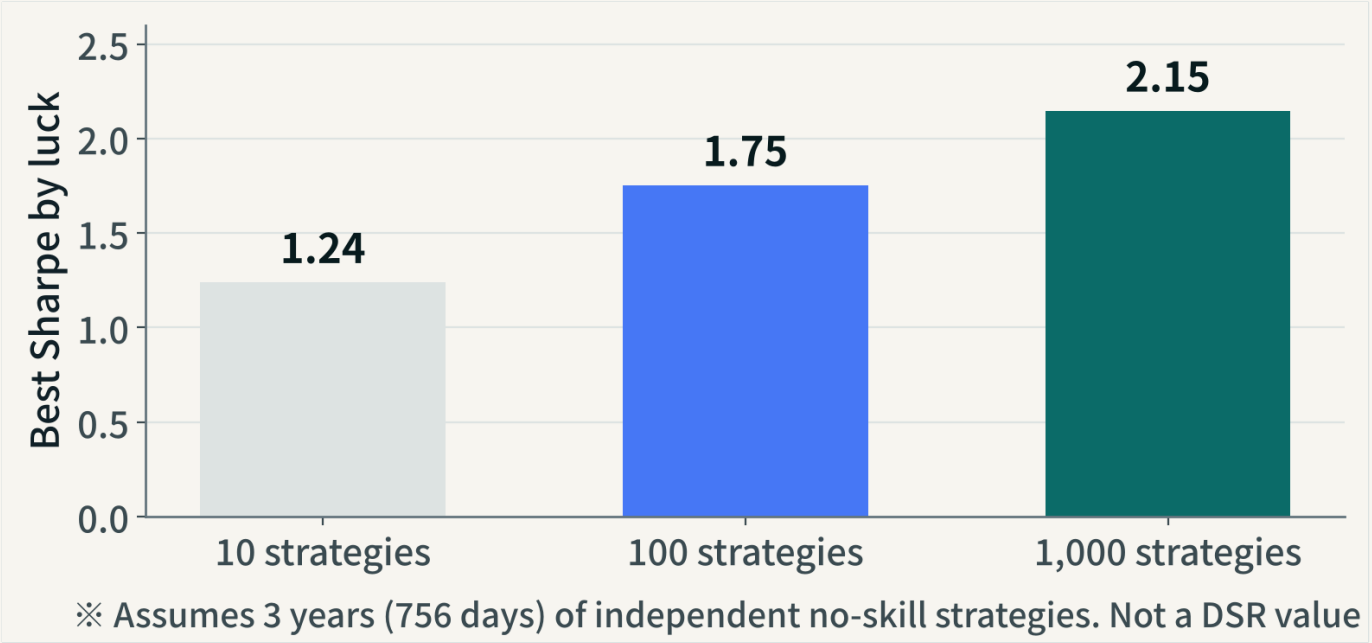
※ Source p.8 divides variance by 5 for teaching. Sample SD (denominator 4) gives another value

Many trials yield good strategies by luck

More trials raise the top Sharpe even with no skill

Multiple testing

- Among 1,000 coin-flip strategies, some win streaks by luck.
- With 1,000 tests at 5%, about 50 look significant even if all are null.



Reporting only the winner overstates skill

The expected 50 needs no independence, but the winner's distribution does

The luck-only top Sharpe is a rough benchmark

1,000 trials on 3 years (756 days) reach about 2.15 with no skill

Source's rough formula

Approx. best Sharpe by luck $\approx \sqrt{252}\,\frac{\sqrt{2\ln N}}{\sqrt{T}}$, $N =$ number of trials, $T =$ number of daily obs

Limits

- An intuitive value for independent no-skill trials; not DSR itself.

DSR (Deflated Sharpe Ratio)

- Judges observed Sharpe given trials, sample length, skew, kurtosis and dependence.

Do not read DSR 0.95 as '95% posterior probability of skill'

03

Lessons from cases

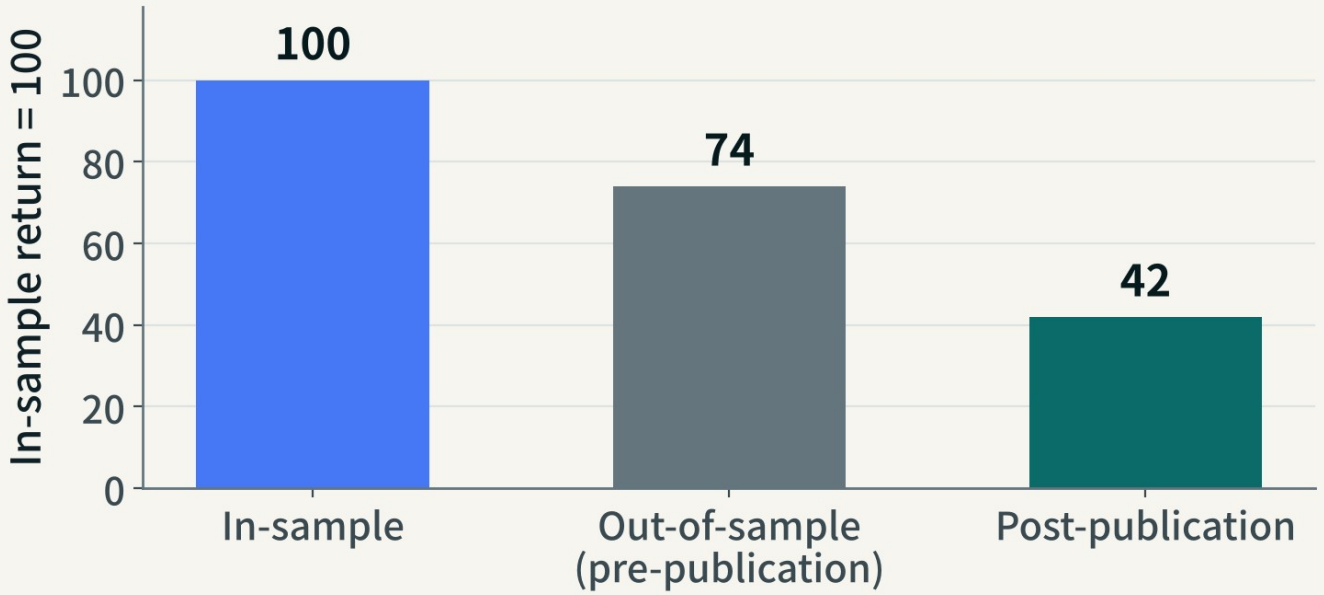
What academia and industry actually observed

Published signals lost much of their return after publication

McLean and Pontiff (2016, Journal of Finance): 97 return predictors tracked

What they found

- Returns fell about 26% after the sample period.
- Returns fell about 58% after publication.



※ McLean & Pontiff (2016), 97 predictors. 26% and 58% declines as index

Partly overfitting, partly investors copying it

Read backtest results as an upper bound on the future

Seeing only survivors makes the past look better

Survivorship bias: failed samples vanish from the data

Bomber story

- In WWII, bullet holes on returning bombers pointed to the wrong spots to armor.
- Planes that never returned left no data.

Fund data

- Dropping closed funds inflates average returns.

Index members

- Using today's members drops delisted stocks.

Backtest on the full universe investable at the time

Hundreds of reported factors: a classic multiple-testing case

Factor zoo: with enough trials, some factors look significant

Harvey·Liu·Zhu (2016)

- Over 300 return factors had been reported.
- They proposed a stricter t ≥ 3.0 hurdle for new factors.

Butter and S&P 500

- Leinweber ‘explained’ the S&P 500 with Bangladesh butter output.
- Data mining finds absurd links.

Lesson

- Record and disclose trial counts.
- Test economically grounded ideas first.

Check not just the winner's score but how many tries it took

04

Summary

Close the week with key Q&A

This week: three checks on information, overlap and luck

One question, one answer for this week

Check question	Answer
Is a time-ordered split enough?	**No. Also check label overlap, preprocessing leakage, test reuse.**
Does high Sharpe prove skill?	**Only after checking trials, data length, costs and leakage.**
Are purging and walk-forward the same?	**No. One removes overlap; the other enforces time order.**
Is DSR 0.95 a 95% skill chance?	**No. It is statistical evidence under stated assumptions.**

Honest validation before fancy models

WHEN DID YOU KNOW IT?

HOW MANY TIMES DID YOU TRY?

THEN TRUST THE SHARPE.

Report decision-time info
and trial counts together.